what does it mean The bandwidth captured by the filter

In signal processing, the "bandwidth captured by the filter" refers to the specific **range of frequencies** that a filter allows to pass through with minimal loss, while blocking or "rejecting" others.

Think of a filter as a specialized "gate" for frequencies. The bandwidth is the "width" of that gate.

Key Components of Filter Bandwidth

- **The Passband**: This is the actual range of frequencies that "squeezes through" the filter. For example, if a filter is designed for a radio station at 100 MHz, its passband might be a small window around that exact frequency.
- **Cutoff Frequencies**: These are the boundaries that define the start and end of the captured bandwidth.
  - **Lower Cutoff (f_L):** The frequency where the filter starts letting signals through.
  - **Upper Cutoff (f_H):** The frequency where the filter starts blocking signals again.
- **The Calculation**: The numerical value for bandwidth (BW) is simply the difference between these two points: $BW = f_H - f_L$.

The "-3 dB" Standard

In technical terms, "captured" doesn't mean the filter suddenly stops working at a specific line. Instead, bandwidth is usually measured at the **-3 dB points**. This is the point where the signal's power has dropped to **half** of its maximum value. Anything within this half-power range is considered "captured" or passed by the filter.

Why It Matters

- **Noise Reduction**: By capturing only the necessary bandwidth for a signal, a filter removes unwanted background "noise" outside that range.
- **Data Rate**: In telecommunications, the wider the bandwidth a filter can capture, the more data (bits per second) it can typically transmit.

Let me know if you're looking for a **specific type of filter** (like low-pass or band-pass) or if you want to see how to **calculate this for a real-world circuit.**

Ask anything

Why it is measured in octaves?

12 sites

Measuring filter bandwidth in **octaves** is standard because it mimics how humans perceive sound and simplifies the design of complex systems. An octave represents a **doubling of frequency** (a 2:1 ratio) rather than a fixed number of Hertz.

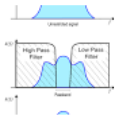
Define the term effective bandwidth of a filter. | Filo
Jan 17, 2026 — Definition of Effective Bandwidth of a filter: The effective bandwidth of a filter is defined as the frequency range within which ...

1. Human Perception (Logarithmic Hearing)

The human ear does not hear frequency linearly. We perceive the "distance" between 100 Hz and 200 Hz as identical to the distance between 1,000 Hz and 2,000 Hz, even though the second jump is 10 times larger in terms of raw Hertz.

Passband - ASCENDO Immersive Audio
Passband - Simple Explanation. The range of frequencies that are allowed to pass through a filter like a low-pass, high-pass, or ...

Passband - Wikipedia
Passband: ... This article needs additional citations for verification. Please help improve this article by adding citations to re...



- **Constant Relative Resolution**: Measuring in octaves provides a meaningful measure of noise power that matches our biological frequency resolution.
- **Critical Bandwidths**: In acoustics, the "critical bandwidth" of human hearing is roughly **1/3 octave**, which is why 1/3-octave filters are the industry standard for noise analysis.

2. Constant Percentage Bandwidth (CPB)

Show all

When you define a filter's bandwidth in octaves, it becomes a **Constant Percentage Bandwidth** filter.

- **Hz vs. Octaves**: A "1-octave" filter at a center frequency of 1,000 Hz has a bandwidth of about 707 Hz. At 10,000 Hz, a 1-octave filter would have a bandwidth of 7,070 Hz.
- **Consistency**: This allows engineers to use the same filter design (topology) across different frequency ranges. For example, a filter with a "Q factor" (quality factor) of 1.41 will always have a 1-octave bandwidth, regardless of whether it's filtering bass or treble.

3. Practical Applications

- **Audio Engineering**: Equalizers (EQs) use octave measurements (e.g., "half-octave EQ") to ensure that adjustments feel "natural" to the listener.
- **Acoustic Standards**: International standards for measuring environmental noise, room acoustics, and hearing protection specify octave or fractional-octave bands (like 1/1, 1/3, or 1/12 octaves) to ensure data is manageable and relevant to human exposure.

These articles explain the logarithmic nature of human hearing and the engineering benefits of measuring filter bandwidth in constant percentage bandwidths:

Deriving Fractional Octave Spectra from the FFT with APx
Dec 17, 2013 — FFT analysis is without question a powerful tool fo...
Audio Precision

Octave-Band Analysis: The Mathematical and Engineeri...
Dec 23, 2025 — Octave-band analysis is a family of spectrum...
CRYSOUND

If you're working on a specific project, I can help you **calculate the Q factor** for a given octave width or explain how **1/3 octave bands** are used in noise dosimetry.

what does it mean "Spatial frequency of the harmonic function. Specified in pixels" relative to gabor filter

In the context of a **Gabor filter**, the "spatial frequency" **defines how many cycles of the wave occur within a certain distance**. When specified in **pixels**, it is describing the **wavelength (λ)** of the underlying sinusoidal (harmonic) component.

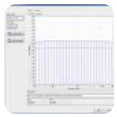
Think of a Gabor filter as a "patch" of stripes. The spatial frequency determines how **thick** or **thin** those stripes are.

14 sites

Deriving Fractional Octave Spectra from the FFT with APx

Dec 17, 2013 — FFT analysis is without question a powerful tool for audio analysis and for signal processing in general. In some applications, no...

Audio Precision



How to Interpret "In Pixels"

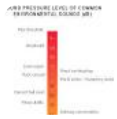
When an algorithm asks for the spatial frequency in pixels, it is usually asking for the **wavelength (λ)**. This represents the distance (in pixels) from one bright stripe to the next.

- **Small Pixel Value (e.g., 5 pixels):** The stripes are very close together. This creates a **high spatial frequency**. This filter is sensitive to fine details, sharp edges, and thin textures.
- **Large Pixel Value (e.g., 30 pixels):** The stripes are wide and spread out. This creates a **low spatial frequency**. This filter ignores fine details and focuses on large, broad shapes or blobs.

Logarithmic scales of frequency and amplitude

Logarithmic Scales of Frequency and Amplitude. We hear the size of the difference in pitch between C' (middle C) and C'' as the same

Physics Stack Exchange



Introduction to Logarithmic Scales | Middle Tennessee State ...

Aug 24, 2023 — Some quick math should convince you that if A above middle C has a frequency of 440 Hz then the lowest A has a frequency of 27.5 Hz

Middle Tennessee State University

The Relationship

The relationship between spatial frequency (f) and wavelength (λ) is inverse:

Show all

$$f = \frac{1}{\lambda}$$

- If $\lambda = 10$ pixels, the frequency is 0.1 cycles per pixel.
- If $\lambda = 2$ pixels (the theoretical limit), the frequency is 0.5 cycles per pixel.

Why the Harmonic Function Matters

A Gabor filter is essentially a **Gaussian "envelope"** (a blurry blob) multiplied by a **Harmonic "carrier"** (a sine or cosine wave).

1. **The Harmonic Function:** Creates the stripes. It tells the filter, "Look for changes in brightness that happen at this specific rhythm."
2. **The Orientation (θ):** Rotates those stripes to look for horizontal, vertical, or diagonal edges.
3. **The Spatial Frequency:** Sets the scale. If you are trying to detect **human hair**, you would use a high spatial frequency (small pixel wavelength). If you are trying to detect

Medium (10–20 px)	Distinct stripes	General patterns, object boundaries.
Large (30+ px)	Broad, soft waves	Global shapes, overall illumination.

Would you like to see how the spatial frequency interacts with the "Sigma" (σ) parameter to change the size of the filter's "search window"?

how it is related to cycles/pixel ratio

In Gabor filtering and digital image processing, **Spatial Frequency** and the **Cycles/Pixel ratio** are two sides of the same coin. They describe the same physical reality using different units of measurement.

Here is the breakdown of how they relate:

1. The Inverse Relationship

The "Spatial Frequency" (f) is the mathematical inverse of the "Wavelength" (λ).

- **Wavelength (λ):** The number of **pixels per cycle** (e.g., "This wave repeats every 10 pixels").
- **Spatial Frequency (f):** The number of **cycles per pixel** (e.g., "In 1 pixel, 0.1 of a wave cycle occurs").

The formula is simply:

$$f = \frac{1}{\lambda}$$

2. Practical Examples

If your Gabor filter documentation asks for spatial frequency, it is asking for the **Cycles/Pixel** value. Here is how the numbers look in practice:

Wavelength (λ) in Pixels	Cycles/Pixel Ratio (f)	Description
2 Pixels	**0.5**	The "Nyquist Limit." The highest possible frequency. Looks like a checkerboard.
4 Pixels	**0.25**	High frequency. Very thin, sharp stripes.
10 Pixels	**0.1**	Medium frequency. Standard texture detection.
100 Pixels	**0.01**	Very low frequency. Broad, slow gradients.

4. Why Use Cycles/Pixel Instead of Just Pixels?

Engineers often prefer the **Cycles/Pixel** ratio because it is "normalized."

When you use a ratio, the math behind the filter (the Fourier Transform) becomes much cleaner. It allows you to describe the filter's behavior in the **Frequency Domain**. For example, if you know your camera sensor produces noise at a frequency of 0.4 cycles/pixel, you can design a Gabor filter at exactly that frequency to "capture" or remove that noise.

Summary

- **Pixels per cycle** (λ) tells you how **wide** one stripe is.
- **Cycles per pixel** (f) tells you how **dense** the stripes are.

Would you like to see how to convert a specific frequency (like "cycles per image width") into the cycles/pixel ratio for your specific image size?